Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/28/2021
PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	06/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 6/30/2023 per PI (Ed., 8/5/21) NOTE: Extended to 9/30/2021 per PI (Ed., 7/8/19) NOTE: Extended to 9/30/2019 per PI (Ed., 7/5/16) NOTE: Extended to 10/1/16 (Ed., 7/31/13) NOTE: To be extended past 10/1/2013 per PI (Ed., 7/24/13)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Johnson, William Ph.D. (California Institute of Technology) Roberts, Scott Ph.D. (Jet Propulsion Laboratory) Bordeenithikasem, Punnathat Ph.D. (NASA Jet Propulsion Laboratory (JPL)/Caltech)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. Amorphous metals (AMs), also known as Bulk Metallic Glasses (BMGs), are a unique class of metal alloys known for their non-crystalline microstructures, unique mechanical properties, and low melting temperatures. BMG matrix composites are amorphous metals reinforced with crystalline phases to improve mechanical properties such as toughness, strength, and wear resistance. When BMGs become crystalline, their toughness and wear resistance is normally severely decreased. By strategically adding crystalline particles, such as dense tungsten (W), within the crystalline BMG matrix, the wear resistance can be remarkably improved. However, sedimentation dramatically affects how these composites can be processed. The microgravity environment aboard the International Space Station (ISS) will be utilized to perform a semi-solid, isothermal (or mushy) hold on the composites between the solidus and the liquidus to selectively precipitate out dendrites into the BMG matrix, resulting in alloys that have wear resistance and toughness that cannot be matched through terrestrial processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
Task Progress:	The research in FY21 produced the following journal publications (Ed. note: see also Bibliography section): 1. V. Hasannaeimi, S. Muskeri, B. Gwalani, D.C. Hofmann, S. Mukherjee. “Deformation behavior of metallic glass composites and plasticity accommodation at microstructural length-scales,” Materials Today Communications, 24, 101237 (2020) 2. P. Bordeenithikasem, S.N. Roberts, D.C. Hofmann, J.M. Ratliff, B.R. Greene, J.B. Bacon, S. Sohn, J. Schroers. “Measuring Demisability of Bulk Metallic Glasses for Potential Satellite Applications through Ablation Experiments,” Adv. Eng. Materials, 2000708 (2020) 3. M. Mohr, D.C. Hofmann, H.-J. Fecht.“Thermophysical Properties of an Fe57.75Ni19.25Mo10C5B8 Glass-Forming Alloy Measured in Microgravity,” Adv. Eng. Materials, 2001143 (2020) In FY21, the team successfully completed the first ground-truth test of the flight experiment to determine if the science objectives are likely to be achieved in the absence of gravity. The ground-truth experiment showed a strong correlation between gravity and the microstructure of the samples and a significant amount of diffusion, demonstrating complete success.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Articles in Peer-reviewed Journals	Hasannaeimi V, Muskeri S, Gwalani B, Hofmann DC, Mukherjee S. "Deformation behavior of metallic glass composites and plasticity accommodation at microstructural length-scales." Materials Today Communications. 2020 Sep;24:101237. https://doi.org/10.1016/j.mtcomm.2020.101237 , Sep-2020
Articles in Peer-reviewed Journals	Bordeenithikasem P, Roberts SN, Hofmann DC, Ratliff JM, Greene BR, Bacon JB, Sohn S, Schroers J. "Measuring demisability of bulk metallic glasses for potential satellite applications through ablation experiments." Advanced Engineering Materials. 2020 Dec;22(12):2000708. https://doi.org/10.1002/adem.202000708 , Dec-2020

Articles in Peer-reviewed Journals	Mohr M, Hofmann DC, Fecht H-J. "Thermophysical properties of an Fe57.75Ni19.25Mo10C5B8 glass-forming alloy measured in microgravity." Advanced Engineering Materials. 2021 Mar;23(3):2001143. https://doi.org/10.1002/adem.202001143 , Mar-2021